

The Method of Undetermined Coefficients:

Defn: Consider a undetermined coefficient (UC) function f . The set of functions consisting of f itself and all linearly independent UC functions of which the successive derivatives of f are either constant multiples or linear combinations will be called the UC set of f .

Ex 1: Solve by using MUCs:

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 2e^x - 10 \sin x \quad \text{--- (1)}$$

Soln: We are given

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 2e^x - 10 \sin x \quad \text{--- (1)}$$

The corresponding homogeneous equation is

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 0 \quad \text{--- (2)}$$

$$\text{A.E. : } m^2 - 2m - 3 = 0$$

$$\Rightarrow (m-3)(m+1) = 0$$

$$\Rightarrow m = 3, -1$$

The complementary f^m is

$$y_c = c_1 e^{3x} + c_2 e^{-x} \quad \text{--- (3)}$$

The nonhomogeneous term is the linear combination of $2e^x - 10 \sin x$ of the two UC functions given by e^x and $\sin x$.

1. Form the UC set for each of these two f's.
we find $S_1 = \{e^x\}$,

$$S_2 = \{\sin x, \cos x\}$$

2. Note that neither of these sets is identical with nor included in the other, hence both are retained.

3. Furthermore, by examining the complementary function, we see that none of the functions $e^x, \sin x, \cos x$ in either of these sets is a soln of the corresponding homogeneous eqn. Hence, neither ~~set~~ set needs to be revised.

4. Thus the original sets S_1 and S_2 remain intact in this problem, and we form the linear combination

$$A e^x + B \sin x + C \cos x$$

of the three elements $e^x, \sin x, \cos x$ of S_1, S_2
~~and~~ with the undetermined coefficients A, B, C

5. We determine these unknown coefficients by taking

$$y_p = A e^x + B \sin x + C \cos x \quad \text{--- (4)}$$

as a particular ~~int~~ solution / ~~integral~~ of (1).

Then

$$y_p' = A e^x + B \cos x - C \sin x$$

$$y_p'' = A e^x + B \sin x + C \cos x$$

Substituting in (1), we find

$$(Ae^x - B \sin x - C \cos x) - 2(Ae^x + B \cos x - C \sin x) - 3(Ae^x + B \sin x + C \cos x) = 2e^x - 10 \sin x$$

Since this is to satisfy the DE (1) hence the coefficients of like terms of both sides must be respectively equal. Equating the coefficients of like terms, we obtain the eqns

$$-4A = 2, \quad -4B + 2C = -10, \quad -4C - 2B = 0$$

solving these equations, we get

$$A = -\frac{1}{2}, \quad B = 2, \quad C = -1$$

and hence we obtain the particular integral

$$y_p = -\frac{1}{2}e^x + 2 \sin x - \cos x$$

Thus the general soln of the given DE (1) is

$$y = y_c + y_p = c_1 e^{2x} + c_2 e^{-x} - \frac{1}{2}e^x + 2 \sin x - \cos x$$

□

Ex 2: Solve $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = 2x^2 + e^x + 2x e^x + 4e^{3x}$ (1)

Soln: the corresponding HE is

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = 0 \quad (2)$$

and the C.F. is $y_c = c_1 e^x + c_2 e^{2x}$ (3)

A.E.
 $m^2 - 3m + 2 = 0$
 $(m-2)(m-1) = 0$
 $m = 1, 2$

The Non-Homogeneous term is the linear combination

$$2x^2 + e^x + 2x e^x + 4e^{3x}$$

of the VC fns given by $x^2, e^x, x e^x, e^{3x}$

1. Form the UC set for each of these f^m s. we have

$$S_1 = \{x^2, x, 1\}$$

$$S_2 = \{e^x\}, S_3 = \{xe^x, e^x\}, S_4 = \{e^{3x}\}$$

2. We note that S_2 is completely included in S_3 , so S_2 is omitted from further consideration, leaving the three sets

$$S_1 = \{x^2, x, 1\}, S_3 = \{xe^x, e^x\}, S_4 = \{e^{3x}\}$$

3. We now observe that $S_3 = \{xe^x, e^x\}$ includes e^x , which is included in the C.F. and so is a soln of the corresponding HDE. Thus we multiply each member of S_3 by x to obtain the revised family

$$S_3' = \{x^2e^x, xe^x\}$$

which ~~obtain~~ contains no member that are soln of the corresponding HE (2).

4) Thus we have the UC sets

$$S_1 = \{x^2, x, 1\}, S_4 = \{e^{3x}\},$$

$$S_3' = \{x^2e^x, xe^x\}$$

we form the linear combination

$$Ax^2 + Bx + C + De^{3x} + Ex^2e^x + Fxe^x$$

of these six elements with ~~un~~ UC A, B, C, D, E, F

5) we determine these unknown by taking a particular soln,

$$y_p = Ax^2 + Bx + C + De^{3x} + Ex^2e^x + Fxe^x$$

from this, we have

$$y_p' = 2Ax + B + 3De^{3x} + Ex^2e^x + 2Exe^x + Fxe^x + Fe^x$$

$$y_p'' = 2A + 9De^{3x} + Ex^2e^x + 4Exe^x + 2Ee^x + Fxe^x + 2Fe^x$$

Substituting y_p, y_p', y_p'' in the DE (1) for y, y', y'' respectively, we obtain

$$\begin{aligned} & 2A + 9De^{3x} + Ex^2e^x + (4E + F)xe^x + (2E + 2F)e^x \\ & - 3[2Ax + B + 3De^{3x} + Ex^2e^x + (2E + F)xe^x + Fe^x] \\ & + 2[Ax^2 + Bx + C + De^{3x} + Ex^2e^x + Fxe^x] \\ & = 2x^2 + e^x + 2xe^x + 4e^{3x} \end{aligned}$$

$$\begin{aligned} \text{or, } & (2A - 3B + 2C) + (2B - 6A)x + 2Ax^2 + 2De^{3x} \\ & + (-2E)xe^x + (2E - F)e^x \\ & = 2x^2 + e^x + 2xe^x + 4e^{3x} \end{aligned}$$

Equating the coefficients of like terms $\Rightarrow$

$$2A - 3B + 2C = 0 \quad \text{Solving these eqns we get}$$

$$2B - 6A = 0$$

$$2A = 2$$

$$2D = 4$$

$$-2E = 2$$

$$2E - F = 1$$

$$A = 1, B = 3, C = 7/2,$$

$$D = 2, E = -1, F = -3$$

So the particular integral is

$$y_p = x^2 + 3x + 7/2 + 2e^{3x} - x^2e^x - 3xe^x$$

The general soln is therefore,

$$y = y_c + y_p = C_1e^x + C_2e^{2x} + x^2 + 3x + 7/2 + 2e^{3x} - x^2e^x - 3xe^x$$

The Method of Undetermined Coefficients:

Def: Consider a undetermined coefficients (UC), ex:
f. the set of fns of which the successive derivatives of f are either

Variati

Solve $\frac{d^4 y}{dx^4} + \frac{d^2 y}{dx^2} = 3x^2 + 4 \sin x - 2 \cos x$

Solve the initial-value Problem

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 2e^x - 10 \sin x$$

$$y(0) = 2$$

$$y'(0) = 4$$

Hint: we have

$$y = c_1 e^{3x} + c_2 e^{-x} - \frac{1}{2} e^x + 2 \sin x - \cos x$$

$$\Rightarrow \frac{dy}{dx} = 3c_1 e^{3x} - c_2 e^{-x} - \frac{1}{2} e^x + 2 \cos x + \sin x$$

Applying I.C $\Rightarrow$

$$2 = c_1 + c_2 - \frac{1}{2} - 1 \Rightarrow c_1 + c_2 = \frac{7}{2}$$

$$4 = 3c_1 - c_2 - \frac{1}{2} + 2 \Rightarrow 3c_1 - c_2 = \frac{5}{2}$$

$$\text{Solving } \Rightarrow c_1 = \frac{3}{2}, c_2 = 2$$

Hence

$$y = \frac{3}{2} e^{3x} + 2 e^{-x} - \frac{1}{2} e^x + 2 \sin x - \cos x$$

□

Variation of Parameters:

Ex: Solve $\frac{d^2 y}{dx^2} + y = \tan x$ --- (1)

solⁿ: the c.f of (1) is $y_c = c_1 \sin x + c_2 \cos x$

We assume $y_p = u_1(x) \sin x + u_2(x) \cos x$ --- (2)

where the functions u_1 and u_2 will be determined such that this is a particular integral of the DE (1). Then

$$y_p' = u_1(x) \cos x - u_2(x) \sin x + u_1'(x) \sin x + u_2'(x) \cos x$$

We impose the condition

$$u_1'(x) \sin x + u_2'(x) \cos x = 0 \quad \text{--- (3)}$$

so that

$$y_p' = u_1(x) \cos x - u_2(x) \sin x$$

$$\Rightarrow y_p'' = -u_1(x) \sin x - u_2(x) \cos x + u_1'(x) \cos x - u_2'(x) \sin x \quad \text{--- (4)}$$

substituting (2) and (4) into (1) we obtain

$$u_1'(x) \cos x - u_2'(x) \sin x = \tan x \quad \text{--- (5)}$$

Thus we have two equations (3) and (5) from which to determine $u_1'(x), u_2'(x)$:

$$\begin{cases} u_1'(x) \sin x + u_2'(x) \cos x = 0 & \text{--- (3)} \\ u_1'(x) \cos x - u_2'(x) \sin x = \tan x & \text{--- (5)} \end{cases}$$

Solving we find:

$$\begin{aligned} u_1'(x) &= \frac{\begin{vmatrix} 0 & \cos x \\ \tan x & -\sin x \end{vmatrix}}{\begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix}} \\ &= \frac{-\cos x \tan x}{-\sin^2 x - \cos^2 x} = \sin x \\ u_2'(x) &= \frac{\begin{vmatrix} \sin x & 0 \\ \cos x & \tan x \end{vmatrix}}{\begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix}} = \\ &= \frac{\sin x \tan x}{-\sin^2 x - \cos^2 x} = -\frac{\sin^2 x}{\cos x} \\ &= \frac{\cos^2 x - 1}{\cos x} = \cos x - \sec x \end{aligned}$$

Integrating, we find

$$\begin{aligned} u_1(x) &= \int \sin x \, dx = -\cos x + c_3 \\ u_2(x) &= \int (\cos x - \sec x) \, dx \\ &= \sin x - \ln |\sec x + \tan x| + c_4 \end{aligned} \quad \text{--- (6)}$$

substituting (6) into (2) we obtain

$$\begin{aligned}
 y_p &= (-\cos x + c_3) \sin x + \sin x + \ln|\sec x + \tan x| + c_4 \cos x \\
 &= -\sin x \cos x + c_3 \sin x + \sin x \cos x \\
 &\quad - \ln|\sec x + \tan x| (\cos x) + c_4 \cos x \\
 &= c_3 \sin x + c_4 \cos x - \cos x (\ln|\sec x + \tan x|)
 \end{aligned}$$

Since the particular integral is a soln free of arbitrary constants, we may assign any ^{particular} values A and B to c_3 and c_4 , respectively, and thus the result will be the P.I.

$$y_p = A \sin x + B \cos x - \cos x (\ln|\sec x + \tan x|)$$

$$\text{Thus } y = y_c + y_p$$

$$\begin{aligned}
 &= c_1 \sin x + c_2 \cos x + A \sin x + B \cos x \\
 &\quad - \cos x (\ln|\sec x + \tan x|)
 \end{aligned}$$

$$\begin{aligned}
 &= (c_1 + A) \sin x + (c_2 + B) \cos x \\
 &\quad - \cos x (\ln|\sec x + \tan x|)
 \end{aligned}$$

$$\Rightarrow y = C_1 \sin x + C_2 \cos x - \cos x (\ln|\sec x + \tan x|)$$

which is the general soln of (1) where
 $C_1 = c_1 + A$ and $C_2 = c_2 + B$.

solve $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = e^x$

by the method of variation of parameters.

$(x^2+1)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 6(x^2+1)^2$

Now x is logarithmic variable with $\cos x$.

from our differential equation to solve
we have ϕ and A solutions and ψ and B solutions
and then the result will be

$(\cos x + \sin x / x) \cos x - \cos x \sin x + \sin x A = 96$

$96 + 96 = 192$

$\cos x B + \sin x A + \cos x C + \sin x D =$

$(\cos x + \sin x / x) \cos x - \cos x \sin x =$

$\cos x (B + C) + \sin x (A + D) =$

$(\cos x + \sin x / x) \cos x - \cos x \sin x + \sin x D = 192$

from (1) to solve however it is difficult
it is a very long process

(6)

The Cauchy - Euler Equation / Homogeneous Linear Equations :

An equation of the form

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = F(x) \dots (1)$$

where $a_0, a_1, a_2, \dots, a_n$ are constants & F is a fⁿ of x , is ~~known~~ called the Cauchy - Euler eqn or equi-dimensional eqn or homogeneous linear equation.

In this eqn, each term in the left member is a constant multiple of an expression of the form $x^k \frac{d^k y}{dx^k}$.

Important substitution / transformation:

The transformation $x = e^t$ reduces the eqn (1) to a linear DE with constant ~~coeff~~ coefficients.

Consider 2nd-order Cauchy - Euler eqn:

$$a_0 x^2 \frac{d^2 y}{dx^2} + a_1 x \frac{dy}{dx} + a_2 y = 0 \text{ or } F(x). \quad (2)$$

Let $x = e^t$, $x > 0$, we have
 $t = \ln x$.

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt} \Rightarrow \frac{dy}{dt} = x \frac{dy}{dx}$$

$$\text{and } \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right)$$

$$= \frac{1}{x} \frac{d}{dx} \left(\frac{dy}{dt} \right) + \frac{dy}{dt} \frac{d}{dx} \left(\frac{1}{x} \right)$$

$$= \frac{1}{x} \frac{d}{dt} \left(\frac{dy}{dt} \right) \frac{dt}{dx} + \frac{dy}{dt} \left(-\frac{1}{x^2} \right)$$

$$= \frac{1}{x} \frac{d^2y}{dt^2} \frac{dt}{dx} - \frac{1}{x^2} \frac{dy}{dt}$$

$$= \frac{1}{x} \left(\frac{d^2y}{dt^2} \right) \cdot \frac{1}{x} - \frac{1}{x^2} \frac{dy}{dt}$$

$$\Rightarrow x^2 \frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt}$$

Substituting in eqn (2) $\Rightarrow$

$$a_0 \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) + a_1 \frac{dy}{dt} + a_2 y = F(t)$$

$$\Rightarrow A_0 \frac{d^2y}{dt^2} + A_1 \frac{dy}{dt} + A_2 y = G(t)$$

where $A_0 = a_0$, $A_1 = a_1 - a_0$, $A_2 = a_2$, $G(t) = F(t)$

The proof in the general n th order case proceeds in a similar fashion.

Ex: Solve $x^2 \frac{dy}{dx} - 2x \frac{dy}{dx} + 2y = x^3$.

[See S.L. Ross]

Ex: $x^3 \frac{d^3y}{dx^3} - 4x^2 \frac{d^2y}{dx^2} + 8x \frac{dy}{dx} - 8y = 4 \ln x$ (1)

Sol: Assuming $x > 0$, we let $x = e^t$, then $t = \ln x$ and

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right)$$

$$= \frac{1}{x} \frac{d^2y}{dt^2} \frac{dt}{dx} + \frac{dy}{dt} \frac{d}{dx} \left(\frac{1}{x} \right)$$

$$= \frac{1}{x^2} \frac{d^2y}{dt^2} - \frac{1}{x^2} \frac{dy}{dt}$$

$$= \frac{1}{x^2} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right)$$

$$\frac{d^3y}{dx^3} = \frac{1}{x^3} \frac{d}{dx} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) + \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) \frac{d}{dx} \left(\frac{1}{x} \right)$$

$$= \frac{1}{x^3} \left(\frac{d^3y}{dt^3} \frac{dt}{dx} - \frac{d^2y}{dt^2} \frac{dt}{dx} \right) - \frac{2}{x^3} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right)$$

$$= \frac{1}{x^3} \left(\frac{d^3y}{dt^3} - 3 \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} \right)$$

Thus, substituting in (1) $\Rightarrow$

$$\left(\frac{d^3y}{dt^3} - 3\frac{d^2y}{dt^2} + 2\frac{dy}{dt}\right) - 4\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right) + 8\frac{dy}{dt}$$

$$- 8y = 4t$$

$$\Rightarrow \frac{d^3y}{dt^3} - 7\frac{d^2y}{dt^2} + 14\frac{dy}{dt} - 8y = 4t \quad (2)$$

The C.F. of (2) is:

$$y_c = c_1 e^t + c_2 e^{2t} + c_3 e^{4t}$$

$$\begin{cases} m^3 - 7m^2 + 14m - 8 = 0 \\ (m-1)(m-2)(m-4) = 0 \\ \Rightarrow m = 1, 2, 4 \end{cases}$$

To find the particular integral of (2) we use method of undetermined coefficients.

$$\text{V.C. set } S = \{t, 1\}$$

We assume $y_p = At + B$ then

$$y_p' = A, \quad y_p'' = y_p''' = 0$$

$$\therefore (2) \Rightarrow 14A - 8(At + B) = 4t$$

$$\Rightarrow -8A = 4 \quad \& \quad 14A - 8B = 0$$

$$\text{and so } A = -\frac{1}{2}, B = -\frac{7}{8}$$

$\therefore$ Thus the P.I. of (2) is

$$y_p = -\frac{1}{2}t - \frac{7}{8}$$

and the general soln of (2) is

$$y = y_c + y_p = c_1 e^t + c_2 e^{2t} + c_3 e^{4t} - \frac{1}{2}t - \frac{7}{8}$$

and hence the g.s of (1) is

$$y = c_1 x + c_2 x^2 + c_3 x^4 - \frac{1}{2} \ln x - \frac{7}{8} \quad \square$$